Clip Socks to a Hanger SOP (1x Episode: 3 Pairs)

One episode matches three pairs of socks from one mixed pile, clips each pair to the sock hanger by its cuffs, spaces the three pairs evenly along the hanger, and hangs the hanger on the rail.

The socks are matched on the table. The left gripper (Config L) or the right gripper (Config R) takes the socks off the pile one at a time. A sock whose mate is not out yet goes in the next free place in the **match row**. A sock whose mate is already out goes on top of its mate, cuff on cuff. When the pile is empty, three pairs lie in the match row.

Each pair is clipped by the left gripper and the right gripper working together. The right gripper picks the pair up by both cuffs and holds it under the clip. The left gripper squeezes the clip open. The right gripper pushes the cuffs up into the open jaws. The left gripper lets go so the clip closes on the cuffs, and then the right gripper lets go.

The order never changes: match all three pairs, clip the left pair to the left clip, the middle pair to the middle clip, the right pair to the right clip, space the pairs, hang the hanger on the rail. Clipping starts only once all three pairs are matched. Spacing starts only once all three pairs are clipped.

The table is set up in one of two ways. Only the sock pile moves; the match row, the hanger stand, the hanger, and the rack are in the same place in both.

- **Config L:** the sock pile is in the front-left corner.
- **Config R:** the sock pile is in the front-right corner.

Where a step depends on the setup it says so on an **IF** line: look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the sock pile is at the front-left (**Config L**) or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Pile-side rule:** in Config L the left gripper takes every sock off the pile and lays it in the match row, and the right gripper waits clear of the pile. In Config R the right gripper takes every sock off the pile and lays it in the match row, and the left gripper waits clear of the pile. No arm reaches across the table to the pile.
- **Fixed roles:** everything after the matching is the same in both configs. The left gripper opens and closes the clips and pushes the left clip along the bar. The right gripper carries each pair, seats the cuffs in the clip, pushes the right clip along the bar, and hangs the hanger on the rail.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the sock pile for this episode's config, the match row, the hanger on its stand, and the rack across the back.
3. The whole bar of the hanger is in frame, from its left end to its right end, so every clip and every cuff can be seen.
4. The rail is in frame end to end.
5. The left arm and the right arm are at home, the left gripper and the right gripper open.
6. The tabletop holds nothing except the sock pile, the hanger stand with the hanger, the rack, and robot hardware. The match row is bare.
7. Nothing blocks the paths between the pile, the match row, the hanger, and the rail.
8. The left arm reaches the front-left corner, the whole match row, and all three clips. The right arm reaches the front-right corner, the whole match row, all three clips, the hanger's neck, and the middle of the rail. Neither arm leans out or stretches to a joint limit to get there.

Materials checklist

1. Six **socks**: three **pairs**. The two socks of a pair are the same colour, pattern, and size. The three pairs are clearly different from each other by colour or pattern.
2. All six socks are the same length, dry, clean, right side out, and lying open. None is balled, rolled, or tucked into another.
3. The six socks lie in one loose **sock pile** in the corner for this episode's config, crossing each other. No pair lies with its two socks together on top of the pile.
    - **Config L:** front-left corner
    - **Config R:** front-right corner
4. One **hanger stand** stands at the center of the table, in front of the rack. It stands firm on its own base and does not rock or slide when a clip is squeezed.
5. One **sock hanger** hangs on the stand by its **hook**, hook opening facing the back, the **bar** running left to right and facing the front edge.
6. Three **clips** ride on the bar, side by side in the middle, the middle clip right under the **neck**, with a gap of about one clip's width between them. All three are closed and empty.

7. Every clip opens fully when its levers are squeezed and snaps shut when they are let go. A clip that sticks or does not close is not staged.
8. The stand is tall enough that a pair hanging from a clip hangs clear of the tabletop.
9. One **garment rack** stands across the back of the table, in the middle of the table's back edge, the **rail** running left to right. The rail is empty. The rack does not rock, slide, or lift when a hanger is hung on it.
10. The **match row**, in the middle of the table just in from the front edge, is bare and has room for three socks laid side by side.

Workspace layout

- **Sock pile:** the front-left corner (**Config L**) or the front-right corner (**Config R**). Only the left gripper reaches the front-left corner and only the right gripper reaches the front-right corner.
- **Match row:** the middle of the table just in from the front edge. The socks are matched here. The left arm and the right arm reach all of it.
- **Hanger stand:** the center of the table, behind the match row. The pairs are clipped and spaced here, with the hanger hanging on the stand.
- **Garment rack:** across the back of the table. The finished hanger goes on the middle of the rail. The rack stays where it is for the whole episode and no gripper ever holds it.

The match row, the stand, and the rack stay the same for every episode; only the sock pile moves with the config.

Arm assignments

- **Left gripper:** in Config L takes every sock off the pile and lays it in the match row. In both configs squeezes each clip open and lets it close, and pushes the left clip along the bar.
- **Right gripper:** in Config R takes every sock off the pile and lays it in the match row. In both configs picks up each pair by its cuffs, seats the cuffs in the clip, pushes the right clip along the bar, and hangs the hanger on the rail.
- While the left gripper or the right gripper takes a sock off the pile, the other one waits clear of the pile and the match row. The left gripper and the right gripper never crowd the same spot.

Vocabulary

- **Left and right:** always as seen from the front edge, for the table, the hanger, and the rail.
- **Sock:** one of the six socks. Each has a **cuff**, the stretchy band at its open end, and a **toe**, the closed end. The **cuff edge** is the very end of the cuff, around the opening.
- **Pair:** two socks that are the same colour, pattern, and size. The other sock of a pair is that sock's **mate**.
- **Top sock:** the sock lying highest on the sock pile, with nothing on top of it.

- **Match row:** three places side by side in the middle of the table, just in from the front edge. Each place holds one sock, and later one pair. A sock in the match row lies flat, cuff toward the back and toe toward the front edge.
- **Next free place:** the empty place in the match row closest to the pile. In Config L the row fills from its left end going right. In Config R it fills from its right end going left.
- **Matched pair:** a sock lying on top of its mate in one place of the match row, cuff on cuff and toe on toe.
- **Left pair, middle pair, right pair:** the matched pairs in the left, middle, and right places of the match row.
- **Sock hanger:** the hanger the socks are clipped to. It has a **hook** on top, a short straight **neck** under the hook, and a straight **bar** running left to right under the neck.
- **Clip:** a spring clip that rides on the bar by a ring at its top and slides along it. It has two **levers** at the top, one at the front and one at the back, and two **jaws** at the bottom. Squeezing the levers together opens the jaws; letting the levers go snaps the jaws shut. The **left clip**, **middle clip**, and **right clip** are named by where they ride on the bar.
- **Open clip:** a clip with its levers squeezed together and its jaws held apart.
- **Seated:** both cuffs of a pair sit deep in the closed jaws, cuff edges in line with each other and straight across the jaws, and the pair hangs straight down from the clip, toes down, one sock on the other.
- **Evenly spaced:** the middle clip stays under the neck. The left clip sits just in from the left end of the bar and the right clip just in from the right end, the same way in from each end. The gap from the left pair to the middle pair is the same as the gap from the middle pair to the right pair.
- **Hanger stand:** an upright on a heavy base with an arm at the top. The hanger hangs on that arm by its hook until Step 4.
- **Lift away:** raise the left gripper or the right gripper, or the item it holds, straight up until there is daylight under it, before anything moves sideways.
- **Carry level:** move the item across at one height, without turning, swinging, or shaking it.
- **Settled:** the item stays put when the left gripper or the right gripper lets go, with nothing sliding, swinging away, or falling.
- **Let-go point:** the left gripper and the right gripper open only at one of these: the left gripper (Config L) or the right gripper (Config R) once a sock lies flat in the match row or on its mate; the left gripper once the right gripper holds the cuffs inside the open jaws; the right gripper once the clip has closed on the cuffs; the right gripper once the hook sits over the rail.

Steps

Run Step 1 six times, once per sock. Then run Step 2 three times, once per pair: the left pair to the left clip, then the middle pair to the middle clip, then the right pair to the right clip. Then run Steps 3, 4, and 5 once.

Only Step 1 depends on the config: in Config L the left gripper takes the socks off the pile, and in Config R the right gripper does. Steps 2 to 5 are the same in both configs.

Any grasp may be retried. If the left gripper or the right gripper slips off, or a sock or a cuff it laid lands in the wrong place, that same gripper opens, takes the same point again, and lays it again. A retry is always made by the gripper the step names. Nothing is ever forced to save a retry.

Step 1: Match the pairs

Goal: all six socks are off the pile and lie as three matched pairs in the match row.

1.1 Take the top sock off the pile

- **IF the pile is at the front-left (Config L):** the **left gripper** pinches the **top sock** at its cuff and lifts it straight up until it hangs free of the pile. The **right gripper** waits clear of the pile and the match row.
- **IF the pile is at the front-right (Config R):** the **right gripper** pinches the **top sock** at its cuff and lifts it straight up until it hangs free of the pile. The **left gripper** waits clear of the pile and the match row.

Take one sock at a time. If a second sock comes up with it, the **left gripper** (Config L) or the **right gripper** (Config R) lays the held sock back on the pile and takes the top sock again.

1.2 Lay the sock in the match row

- **IF Config L, and the sock's mate is not in the match row yet:** the **left gripper** carries the sock level to the **next free place**. It lays the sock flat there, toe down first at the front, cuff toward the back. Then the **left gripper** opens and lifts away.
- **IF Config L, and the sock's mate is already in the match row:** the **left gripper** carries the sock level to its mate. It lays the sock flat on top of its mate, cuff on cuff and toe on toe. Then the **left gripper** opens and lifts away.
- **IF Config R, and the sock's mate is not in the match row yet:** the **right gripper** carries the sock level to the **next free place**. It lays the sock flat there, toe down first at the front, cuff toward the back. Then the **right gripper** opens and lifts away.
- **IF Config R, and the sock's mate is already in the match row:** the **right gripper** carries the sock level to its mate. It lays the sock flat on top of its mate, cuff on cuff and toe on toe. Then the **right gripper** opens and lifts away.
- Go back to 1.1 for the next sock. After the sixth sock, go on to Step 2.

Check: every sock in the match row lies flat, cuff toward the back. Every sock laid on another lies on its own mate, cuff on cuff and toe on toe. No place holds two socks of different pairs. After the sixth sock the pile is empty and the match row holds three matched pairs.

If a sock was laid on a sock that is not its mate, the **left gripper** (Config L) or the **right gripper** (Config R) lifts it off by the cuff and lays it in the right place before the next sock comes off the pile. If the cuffs of a matched pair are not on each other, the same gripper lifts the top sock by its cuff and lays it again.

Step 2: Clip a pair to its clip

Goal: the pair hangs from its clip with both cuffs **seated**. Work the left pair on the left clip first, then the middle pair on the middle clip, then the right pair on the right clip.

2.1 Pick up the pair

- The **right gripper** pinches both cuffs of the pair together, in the middle of the cuffs, just below the cuff edges, leaving a strip of cuff above its fingers.
- The **right gripper** lifts the pair straight up until both socks hang toe down, one on the other.
- The **right gripper** carries the pair level to just under the clip, cuff edges just below the jaws.
- The **left gripper** waits clear of the hanger.

2.2 Open the clip

- The **left gripper** pinches the clip's two levers together, one finger on the front lever and one on the back lever, until the jaws stand fully open.
- The **left gripper** holds the levers squeezed and keeps the clip still on the bar.
- The **right gripper** holds the pair still under the clip.

Never open a clip by pulling its jaws apart, and never open it with the right gripper.

2.3 Seat the cuffs

- The **right gripper** raises the pair straight up so both cuff edges go in between the open jaws.
- The **right gripper** keeps raising the pair until the cuff edges sit deep in the jaws, straight across, in line with each other.
- The **left gripper** keeps the levers squeezed and the clip still.

2.4 Let the clip close

- The **left gripper** lets go of the levers slowly so the jaws close on both cuffs, then lifts away.
- The **right gripper** holds the pair still until the jaws have closed.
- The **right gripper** opens and lowers away from the pair.
- Go back to 2.1 for the next pair. After the right pair, go on to Step 3.

Check: both cuffs are **seated**: deep in the closed jaws, straight across, and the pair hangs straight down, toes down, one sock on the other, and **settled**. The hanger still hangs on the stand.

If only one cuff is in the jaws, a cuff sits only by its corner, or the cuffs sit crooked, the **left gripper** squeezes the clip open again, the **right gripper** takes the pair by both cuffs and seats them again, and the **left gripper** lets the clip close.

Step 3: Space the pairs evenly

Goal: the three pairs are **evenly spaced** along the bar.

- The **left gripper**, closed, pushes the left clip by the side of its ring, slowly along the bar to the left, until the left clip sits just in from the left end of the bar.
- The **right gripper**, closed, pushes the right clip by the side of its ring, slowly along the bar to the right, until the right clip sits just in from the right end of the bar.
- The middle clip stays under the neck. The **left gripper** and the **right gripper** lift away.

Push the clip by its ring only. Never squeeze the levers to move a clip, never pull a clip by its socks, and never push a clip off the end of the bar.

Check: the pairs are evenly spaced: the left clip and the right clip sit the same way in from their ends, and the two gaps between the pairs look the same. All three pairs are still seated and hang straight down. The hanger still hangs on the stand.

If the gaps look different, the **left gripper** pushes the left clip, or the **right gripper** pushes the right clip, a little more along the bar until they match.

Step 4: Hang the hanger on the rail

Goal: the hanger hangs on the middle of the rail by its hook, hook opening facing the back, with the three pairs hanging from it.

- The **right gripper** pinches the hanger at its **neck**, just below the hook.
- The **right gripper** lifts the hanger straight up until the hook clears the arm of the stand.
- The **right gripper** carries the hanger level and slowly back toward the rack, without turning it, so the pairs do not swing into the stand or the rack.
- The **right gripper** raises the hook over the middle of the rail and brings it down until the hook sits on the rail, hook opening facing the back.
- The **right gripper** opens and lifts away.
- The **left gripper** waits clear of the hanger and the rack the whole step.

Never lift the hanger by the bar, a clip, or a sock, and never drop the hook onto the rail.

Check: the hook sits over the rail in the middle, hook opening facing the back, the bar facing the front edge. The hanger hangs still and **settled**. All three pairs are still seated and evenly

spaced.

If the hook landed off the rail or turned, the **right gripper** takes the neck again, lifts the hook off, and hangs it again.

Step 5: End the episode

Goal: recording ends with the hanger on the rail and three matched pairs clipped and evenly spaced on it.

- Look once across the table: the hanger on the middle of the rail, three matched pairs seated and evenly spaced, the pile and the match row bare, and the stand empty.
- Return the left arm and the right arm home, the **left gripper** and the **right gripper** open. Homing is the last thing the arms do.
- Stop recording.

Check: the left arm and the right arm are at home, the **left gripper** and the **right gripper** are open, and neither holds anything.

Expected state: the hanger and the pairs are still, the rack and the stand have not moved, and the recording has stopped.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the hanger off the rail and unclip all three pairs.
2. Check every clip still opens fully and snaps shut. Check the hook and the bar are not bent.
3. Hang the hanger back on the stand, hook opening facing the back, bar facing the front edge.
4. Slide the three clips back together in the middle of the bar, the middle clip under the neck, a gap of about one clip's width between them, all three closed and empty.
5. Smooth each sock out, right side out, and lay all six in one loose pile, crossing each other, in the corner for the next episode's config: front-left (Config L) or front-right (Config R). Make sure no pair lies with its two socks together on top.
6. Check the stand stands firm and the rack has not moved.
7. Replace anything damaged: a sock that is torn, stained, damp, or has a stretched cuff; a clip that sticks or will not close; a bent hanger.
8. Clear the match row.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Keep the episode with its violation tags. Do not delete it just because a rule was broken.

Tag a violation only when something is forced, the order is broken, or the episode moves on with a fault left in place.

Violations

Violation: Config misaligned

- **Visible cue:** what the arms do does not match the config on the table: the right gripper takes a sock off the pile in Config L, or the left gripper in Config R; the left gripper or the right gripper reaches across the table to the pile; or the wrong IF line is followed.
- **SOP rule broken:** Step 1, the pile-side rule (the left gripper in Config L and the right gripper in Config R take every sock off the pile, no arm reaches across the table to the pile, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the pile is before the first sock, then follow that config's IF line in Step 1.

Violation: Wrong order

- **Visible cue:** a pair is picked up for clipping before all six socks are off the pile; a clip is pushed along the bar before all three pairs are clipped; or the hanger is lifted off the stand before the pairs are spaced.
- **SOP rule broken:** Steps 1 to 4, match all three pairs, clip all three pairs, space the pairs, then hang the hanger on the rail, in that order.
- **Coaching note:** finish each stage for all three pairs before you start the next.

Violation: Sock taken off the pile the wrong way

- **Visible cue:** a sock other than the top sock is pulled out from under the others; two socks are carried at once; a sock is dragged across the table instead of lifted; or a sock is taken by its toe or the middle of its foot instead of its cuff.
- **SOP rule broken:** Step 1.1, the left gripper (Config L) or the right gripper (Config R) pinches the top sock at its cuff and lifts it straight up, one sock at a time.
- **Coaching note:** top sock only, by the cuff, straight up.

Violation: Socks of different pairs put together

- **Visible cue:** a sock is laid on a sock that is not its mate, or two socks of different pairs are clipped together, and the episode moves on.
- **SOP rule broken:** Steps 1.2 and 2.1, every sock is laid on its own mate, and a sock laid on the wrong one is lifted off and laid in the right place before the next sock comes off the pile.
- **Coaching note:** look at the colour and the pattern before you lay it down. If it is wrong, fix it before the next sock.

Violation: Sock laid out of place in the match row

- **Visible cue:** a sock is laid somewhere other than the next free place or its mate; it is laid outside the match row; it is laid with its cuff toward the front or a side; or it is dropped or laid bunched instead of flat.
- **SOP rule broken:** Step 1.2, a sock with no mate out yet is laid flat in the next free place, cuff toward the back, toe toward the front edge.
- **Coaching note:** next free place, toe toward you, cuff to the back, laid flat.

Violation: Mate not laid cuff on cuff

- **Visible cue:** a mate is laid on its pair with its cuff over the other sock's toe, turned across it, or hanging off to one side, and the episode moves on.
- **SOP rule broken:** Step 1.2, a mate is laid flat on top of its pair, cuff on cuff and toe on toe; if the cuffs are not on each other, the top sock is lifted and laid again.
- **Coaching note:** cuff on cuff, toe on toe. If they are off, lift the top sock and lay it again.

Violation: Pair picked up the wrong way

- **Visible cue:** the right gripper picks up only one sock of the pair; it takes the pair by the toes or the middle of the foot; it pinches right at the cuff edges, leaving no cuff above its fingers; or the pair is dragged across the table to the hanger.
- **SOP rule broken:** Step 2.1, the right gripper pinches both cuffs together in the middle, just below the cuff edges, and lifts the pair straight up so it hangs toe down.
- **Coaching note:** both cuffs, just below the edge, then straight up.

Violation: Pairs clipped to the wrong clip or out of order

- **Visible cue:** a pair goes to a clip other than its own; the middle or the right pair is clipped before the left pair; two pairs go in one clip; or a clip is left empty.
- **SOP rule broken:** Step 2, the left pair goes on the left clip first, then the middle pair on the middle clip, then the right pair on the right clip, one pair per clip.
- **Coaching note:** left, middle, right. One pair per clip.

Violation: Clip opened the wrong way

- **Visible cue:** the clip is opened by pulling its jaws apart, by the right gripper, by pushing the cuffs into the closed jaws, or by squeezing only one lever against the bar; or the jaws are only half open when the cuffs go in.
- **SOP rule broken:** Step 2.2, the left gripper pinches the front lever and the back lever together until the jaws stand fully open, and holds them there.
- **Coaching note:** left gripper, both levers, fully open before the cuffs go in.

Violation: Cuffs not seated

- **Visible cue:** only one cuff is in the jaws; a cuff is held only by its corner or by the very edge; the cuffs sit crooked across the jaws; or the pair hangs twisted, and the episode moves on to the next pair.
- **SOP rule broken:** Steps 2.3 and 2.4, both cuff edges sit deep in the jaws, straight across and in line, and cuffs that are not seated are seated again before the next pair.
- **Coaching note:** both cuffs, deep and straight. If not, open the clip and seat them again.

Violation: Let go in the wrong order

- **Visible cue:** the right gripper opens before the clip has closed on the cuffs, so the pair falls or slips down; the left gripper lets the levers snap shut before the cuffs are deep in the jaws; or the clip snaps shut on a finger of the right gripper.
- **SOP rule broken:** Step 2.4, the left gripper lets go of the levers slowly once the cuffs are deep in the jaws, and the right gripper opens only after the jaws have closed.
- **Coaching note:** cuffs in, left gripper lets go slowly, then the right gripper lets go.

Violation: Clips spaced unevenly

- **Visible cue:** the left clip and the right clip sit different ways in from their ends, the two gaps between the pairs look clearly different, an outer clip is left in the middle, or the middle clip is moved off the neck, and the episode moves on to the rail.
- **SOP rule broken:** Step 3, the left clip goes just in from the left end and the right clip just in from the right end, the middle clip stays under the neck, and the two gaps are made to match before moving on.
- **Coaching note:** look at both gaps before you lift the hanger. Push a clip a little more until they match.

Violation: Clip moved the wrong way

- **Visible cue:** a clip is moved by squeezing its levers, by pulling its socks, or by lifting it off the bar; a clip is pushed off the end of the bar; or the right gripper pushes the left clip or the left gripper the right clip.
- **SOP rule broken:** Step 3, the left gripper pushes the left clip and the right gripper pushes the right clip, each closed and against the side of the ring, slowly along the bar.

- **Coaching note:** closed gripper on the ring, slow push. Left clip with the left gripper, right clip with the right gripper.

Violation: Hanger carried the wrong way

- **Visible cue:** the hanger is lifted by the bar, a clip, or a sock instead of the neck; it is dragged off the stand arm; it swings so the pairs hit the stand or the rack; it is turned on the way; or the left gripper takes part in the carry.
- **SOP rule broken:** Step 4, the right gripper pinches the neck, lifts the hook straight up clear of the stand arm, and carries the hanger level and slowly to the rail without turning it.
- **Coaching note:** neck only, straight up off the stand, slow and level.

Violation: Hanger hung wrong on the rail

- **Visible cue:** the hook is left off the rail, caught on the rail end or a rack upright, hung far from the middle of the rail, or with its opening facing the front; or the hanger is dropped onto the rail instead of brought down onto it, and the episode moves on.
- **SOP rule broken:** Step 4, the right gripper brings the hook down onto the middle of the rail, hook opening facing the back, and opens only once the hook sits over the rail; a hook that landed wrong is lifted and hung again.
- **Coaching note:** hook over the middle, opening to the back, bring it down, then let go.

Violation: Pair knocked off a clip

- **Visible cue:** a pair, or one sock of a pair, slips out of its clip during clipping, spacing, or the carry, and the episode moves on without seating it again.
- **SOP rule broken:** Steps 2 to 4, every pair stays seated in its clip from the moment its clip closes until the end of the episode.
- **Coaching note:** move slowly once a pair is on. If one falls, seat it again before going on.

Violation: Hanger stand moved or knocked

- **Visible cue:** the stand rocks, slides, or tips; the hanger is lifted off the stand arm or swings off it before Step 4; or the left gripper or the right gripper holds or leans on the stand.
- **SOP rule broken:** Steps 2 and 3, the hanger hangs on the stand through the clipping and the spacing, and neither the left gripper nor the right gripper touches the stand.
- **Coaching note:** the stand stays put. Work the clips, not the stand.

Violation: Rack moved or lifted

- **Visible cue:** the rack rocks, slides, tips, or lifts off the tabletop; the rail bends under the hook; or the left gripper or the right gripper holds, drags, or leans on the rack.
- **SOP rule broken:** Step 4, the rack stands on its own base and never moves, and neither the left gripper nor the right gripper touches it.

- **Coaching note:** the rack stands on its own. If it is moving, stop and set it back.

Violation: Arms swapped roles or crowded the same spot

- **Visible cue:** the right gripper squeezes a clip open; the left gripper picks up a pair or seats cuffs; the gripper not taking socks off the pile works over the pile or the match row in Step 1; or the left gripper and the right gripper work the same spot, touch each other, or one has to move before the other can reach.
- **SOP rule broken:** Steps 1 to 4, in Step 1 only the left gripper (Config L) or the right gripper (Config R) works the pile and the match row; the left gripper opens and closes the clips; the right gripper carries the pairs, seats the cuffs, and hangs the hanger.
- **Coaching note:** left gripper opens the clip, right gripper brings the socks. Neither one does the other's work.

Violation: Item dropped or something knocked over

- **Visible cue:** a sock falls onto the tabletop or the floor away from the match row; the pile or the match row is knocked out of place; the hanger falls; or the two arms strike each other, the stand, or the rack.
- **SOP rule broken:** Steps 1 to 4, each item stays on a clear path, and the left gripper and the right gripper open only at a let-go point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the hanger off the rail, fewer than three pairs clipped, a sock still on the table, or the pairs not evenly spaced; or either arm is away from home, the left gripper or the right gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 5, look once across the table, return the left arm and the right arm home with the left gripper and the right gripper open, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Bad clip:** a clip that sticks, will not open fully, will not close, or comes off the bar. Replace it before the next episode.
- **Bad sock:** a torn sock, a stretched cuff that will not stay in the jaws, or two socks of a pair that do not match. Replace it before the next episode.

- **Bad hanger or stand:** a bent hook or bar, or a stand that will not stand firm.
- **Rack fault:** the rack will not stand straight or the rail bends. Repair or replace it before the next episode.

Annotation subtasks (from SOP)

1. Take the top sock off the pile
2. Lay the sock in the match row or on its mate
3. Pick up a pair by its cuffs
4. Squeeze the clip open
5. Seat the cuffs in the clip
6. Let the clip close and let go of the pair
7. Push the left clip and the right clip out to space the pairs
8. Lift the hanger off the stand by its neck
9. Hang the hanger on the rail
10. Return the left arm and the right arm home and end the episode